Phu Nguyen

For this lab I went through the notebook in order and ran the code myself so I could see what each part was doing. I loaded the Titanic data and checked the missing values first, then filled in the missing Age values with the median and made sure none were left. After that I one-hot encoded Sex and Embarked with pd.get_dummies and scaled Age and Fare with StandardScaler.

For the three tasks and the knowledge check questions, I got help from Google Colab assistance. After that I ran the code, looked at the new columns and scaled numbers myself, and wrote the answers so they made sense to me.
This lab made me see why you have to clean the data before putting it in a model. When I finished, I uploaded it to my GitHub repo and submitted it.

Cyndy Ndong

This contribution journal documents the work completed for Lab 05 on Data Preprocessing and

Machine Learning Workflow for course ITAI1371. Every part of this lab was completed step-by-step following the guided notebook instructions.

I started by loading the Titanic dataset directly into Jupyter Lab using Pandas and inspecting the initial missing values across all columns. To clean the missing data, I calculated the median of the Age column and used the fillna method to impute the missing values without letting extreme outliers skew the dataset.
Next, I handled categorical variables by using One-Hot Encoding through Pandas get_dummies

on the Sex and Embarked columns, dropping the first category to avoid multicollinearity and converting text categories into binary indicators that machine learning models can read. I then applied feature scaling to the Age and Fare columns using Scikit-Learn's StandardScaler so features with larger ranges would not dominate model calculations.

Finally, I answered all the embedded Knowledge Check questions in the notebook, explaining the logic behind median imputation, One-Hot Encoding, and why decision trees do not require feature scaling. I verified that every code cell ran smoothly from top to bottom without errors and exported the complete notebook into a PDF deliverable following the exact file naming rules.